

Spring 2021 Real Analysis II — Homework 6

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Unless otherwise specified, $C([a, b])$ is endowed with norm $\|x\| = \max_{a \leq t \leq b} |x(t)|$ for any $x \in C([a, b])$.

1. Let X be a B^* space and X_0 a closed linear subspace. Show that $d(x, X_0) = \sup\{|f(x)| : f \in X^*, \|f\| = 1, f|_{X_0} = 0\}$.

Proof. Denote $E := \{f \in X^* : \|f\| = 1, f|_{X_0} = 0\} \subset \mathbb{R}$ and $s := \sup_{f \in E} |f(x)|$. We need to show $d(x, X_0) = s$.

We know there exists $f \in E$ such that $f(x) = d(x, X_0)$. Hence $s \geq d(x, X_0)$. On the other hand, for any $f \in E$ and $z \in X_0$, we know

$$|f(x)| = |f(x) - f(z)| \leq \|f\| \|x - z\| = \|x - z\|.$$

Taking infimum with respect to z yields $|f(x)| \leq d(x, X_0)$. Hence $s \leq d(x, X_0)$. $\square$

2. Let X be a B^* space. Suppose there are n linearly independent vectors $\{x_i : i = 1, \dots, n\}$ in X , n constants $\{c_i : i = 1, \dots, n\} \subset \mathbb{R}$, and $M > 0$. Show that the following two statements are equivalent:

- There exists $f \in X^*$ such that $f(x_i) = c_i$ for all i and $\|f\| \leq M$.
- For any $a_i \in \mathbb{R}$ there is $|\sum_{i=1}^n a_i c_i| \leq M \|\sum_{i=1}^n a_i x_i\|$.

Proof. $(\Rightarrow)$ We have $|\sum_{i=1}^n a_i c_i| = |f(\sum_{i=1}^n a_i x_i)| \leq \|f\| \|\sum_{i=1}^n a_i x_i\| \leq M \|\sum_{i=1}^n a_i x_i\|$.

$(\Leftarrow)$ Let $X_0 := \text{span}(x_1, \dots, x_n)$ and define $f_0 : X_0 \rightarrow \mathbb{R}$ by $f_0(\sum_{i=1}^n a_i x_i) = \sum_{i=1}^n a_i c_i$ for any $a_i \in \mathbb{R}$. Then it is clear that f_0 is linear, $|f_0(x)| \leq M \|x\| =: p(x)$, and $\|f_0\| \leq M$. Hence, by Hahn-Banach theorem, we can extend $f_0 \in X_0^*$ to $f \in X^*$ such that $f|_{X_0} = f_0$ and $\|f\| = \|f_0\| \leq M$. This also implies $f(x_i) = c_i$ for all i . $\square$

3. Let X be a B^* space and $x_1, \dots, x_n$ be independent vectors in X . Show that there exist $f_1, \dots, f_n \in X^*$ such that $f_i(x_j) = \delta_{ij}$ for all $i, j = 1, \dots, n$.

Proof. For each $i = 1, \dots, n$, denote $X_i := \text{span}(x_1, \dots, \widehat{x_i}, \dots, x_n)$, then $x_i \notin X_i$ and there exists $f_i \in X^*$ such that $f_i(x_i) = 1$ and $f_i|_{X_i} = 0$. It is easy to verify that $f_i(x_j) = \delta_{ij}$ for all $i, j = 1, \dots, n$. $\square$

4. Let $C_0 := \{\xi = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{R}, \lim_{k \rightarrow \infty} |\xi_k| = 0\}$ and define its norm $\|\xi\| := \sup_{k \geq 1} |\xi_k|$. Show that $C_0^* = l_1$.

Proof. ($\subset$) For any $f \in C_0^*$, define $x = (x_1, x_2, \dots)$ such that $x_k = f(e_k)$ for all $k \in \mathbb{N}$. Then we have $f(\xi) = \sum_{k=1}^{\infty} x_k \xi_k$ for any $\xi \in C_0$. So we identify f with x . If $x \notin l^1$, then there exists $\{k_j : j \in \mathbb{N}\}$ such that $\sum_{i=1}^{k_j} |x_i| \geq j$ for every $j \in \mathbb{N}$. Now for any $j \in \mathbb{N}$, define

$$s_i^{(j)} = \begin{cases} \text{sign}(x_i), & i \leq k_j, \\ 0, & i > k_j, \end{cases}$$

then $s^{(j)} = (s_1^{(j)}, s_2^{(j)}, \dots) \in C_0$ and $\|s^{(j)}\| = 1$, but

$$f(s^{(j)}) = \sum_{i=1}^{\infty} s_i^{(j)} x_i = \sum_{i=1}^{k_j} |x_i| \geq j,$$

which contradicts to $f \in C_0^*$. So $x \in l^1$, i.e., $f \in l^1$.

($\supset$) For any $x \in l^1$, identify $f_x \in C_0^*$ defined by $f_x(\xi) = \sum_{k=1}^{\infty} x_k \xi_k$ with x . Then f is linear and $|f_x(\xi)| \leq \|x\|_1 \|\xi\|$. Hence $f_x \in C_0^*$. $\square$

5. Let $x_k, x \in C([a, b])$ for all $k \in \mathbb{N}$ and $x_k \rightharpoonup x$. Show that $\lim_{k \rightarrow \infty} x_k(t) = x(t)$ for all $t \in [a, b]$, i.e., $x_k \rightarrow x$ pointwisely.

Proof. For any $t \in [0, 1]$, define $f_t : C([a, b]) \rightarrow \mathbb{R}$ such that $f_t(x) = x(t)$ for any $x \in C([a, b])$. Then f_t is linear and $|f_t(x)| = |x(t)| \leq \|x\|$, so $f_t \in C([a, b])^*$. Hence

$$\lim_{k \rightarrow \infty} x_k(t) = \lim_{k \rightarrow \infty} f_t(x_k) = f_t(x) = x(t)$$

since $x_k \rightharpoonup x$. $\square$

6. Let X be a B^* space and $x_k \rightharpoonup x_0$. Show that $\liminf_{k \rightarrow \infty} \|x_k\| \geq \|x_0\|$.

Proof. We know there exists $f \in X^*$ such that $f(x_0) = \|x_0\|$ and $\|f\| = 1$. Hence $|f(x_k)| \leq \|f\| \|x_k\| = \|x_k\|$. Taking lower limits of both sides yields $|f(x_0)| = \lim_{k \rightarrow \infty} |f(x_k)| \leq \liminf_{k \rightarrow \infty} \|x_k\|$. $\square$

Alternative proof:

Proof. If not, then $\|x_0\| > 0$, and there exist $\epsilon_0 > 0$ and subsequence $\{x_{k_j}\}$ such that $\|x_{k_j}\| \leq \|x_0\| - \epsilon_0 =: r$ for all $j \in \mathbb{N}$. Since $\bar{B}(0; r)$ is closed and convex in X , we know there exists $f \in X^*$ and $a \in \mathbb{R}$ such that $f(x_{k_j}) < a < f(x_0)$ for all $j \in \mathbb{N}$, which contradicts to $x_k \rightharpoonup x_0$. $\square$

7. Let X be a separable Hilbert space and $\{e_k : k \in \mathbb{N}\}$ a basis (complete orthonormal set) of X . Show that $x_n \rightharpoonup x_0$ if and only if both of the following statements hold:

- There exists $M > 0$ such that $\|x_n\| \leq M$ for all $n \in \mathbb{N}$.
- For any $k \in \mathbb{N}$, there is $\lim_{n \rightarrow \infty} \langle x_n, e_k \rangle = \langle x_0, e_k \rangle$.

Proof. ($\Rightarrow$) We only need to show that $\|x_k\| \leq M$ for some $M > 0$. We identify x_k with its canonical map. Then $\langle x_k, f \rangle = \langle f, x_k \rangle \rightarrow \langle f, x_0 \rangle$ for all $f \in X^*$. Hence $\sup_{k \geq 1} \langle x_k, f \rangle < \infty$. By Uniform Boundedness Theorem, we know $\|x_k\| \leq M$ for some $M > 0$.

($\Leftarrow$) Let $E := \{\sum_{k=1}^K a_k e_k : a_k \in \mathbb{R}, k \in \mathbb{N}\}$. Then $\bar{E} = X = X^*$, i.e., E is dense in X . It is clear that $\langle x_k, f \rangle \rightarrow \langle x, f \rangle$ for all $f \in E$. By Banach-Steinhaus theorem and $\|x_k\| \leq M$, we know $\langle x_k, f \rangle \rightarrow \langle x, f \rangle$ for all $f \in X$. $\square$

8. Let $p \in (1, \infty)$ and $X = L^p(\mathbb{R})$ be endowed with the L^p norm $\|\cdot\|_p$. Define $T_k \in L(X)$ by $(T_k x)(t) = x(t+k)$ for any $t \in \mathbb{R}$, $x \in X$, and $k \in \mathbb{N}$. Show that $T_k \rightarrow 0$ but $\|T_k x\|_p = \|x\|_p$ for all $k \in \mathbb{N}$. [Hint: for any $\epsilon > 0$ and $v \in L^q(\mathbb{R})$, show that both $\int_{-\infty}^{-a} v(t)u(t+k) dt$ and $\int_{-a}^{\infty} v(t)u(t+k) dt$ can be bounded by $\epsilon/2$ for sufficiently large a and k .]

Proof. It is clear that $\|T_k x\|_p = \|x\|_p$ for all $k \in \mathbb{N}$. Note that $X^* = L^q(\mathbb{R})$. Hence, for any $f \in X^*$, there exists $v \in L^q(\mathbb{R})$ such that $f(u) = \int_{-\infty}^{\infty} uv dx$ for all $u \in X$.

Now for any $\epsilon > 0$, $f \in X^*$, and $x \in X$, we know there exists $a = a(v, x, \epsilon) > 0$ (v is the one identified with f) sufficiently large, such that for any $k \in \mathbb{N}$ there is

$$\left| \int_{-\infty}^{-a} v(t)x(t+k) dt \right| \leq \|x\|_p \left(\int_{-\infty}^{-a} |v|^q dx \right)^{1/q} < \frac{\epsilon}{2}.$$

Moreover, there exists $K = K(a, \epsilon) \in \mathbb{N}$, such that for all $k \geq K$ there is

$$\left| \int_{-a}^{\infty} v(t)x(t+k) dt \right| \leq \|v\|_q \left(\int_{-a+k}^{\infty} |x(t)|^p dx \right)^{1/p} < \frac{\epsilon}{2}, \quad \forall k \in \mathbb{N}.$$

Therefore

$$|f(T_k x)| = \left| \int_{-\infty}^{\infty} v(t)x(t+k) dt \right| \leq \left| \int_{-\infty}^{-a} v(t)x(t+k) dt \right| + \left| \int_{-a}^{\infty} v(t)x(t+k) dt \right| < \epsilon$$

for all $k \geq K$. Hence $T_k \rightarrow 0$. □

9. Let X be a Hilbert space, $x_k \rightarrow x_0$, and $y_k \rightarrow y_0$. Show that $\langle x_k, y_k \rangle \rightarrow \langle x_0, y_0 \rangle$.

Proof. Since $x_k \rightarrow x_0$, we know there exists $M > 0$ such that $\|x_k\| \leq M$ for all $k \in \mathbb{N}$.

Since $y_k \rightarrow y_0$ and $x_k \rightarrow x_0$, we know for any $\epsilon > 0$ there exists $K \in \mathbb{N}$ such that $\|y_k - y_0\| \leq \epsilon/(2M)$ and $|\langle x_k, y_0 \rangle - \langle x_0, y_0 \rangle| < \epsilon/2$ for all $k \in \mathbb{N}$. Hence

$$\begin{aligned} |\langle x_k, y_k \rangle - \langle x_0, y_0 \rangle| &\leq |\langle x_k, y_k \rangle - \langle x_k, y_0 \rangle| + |\langle x_k, y_0 \rangle - \langle x_0, y_0 \rangle| \\ &\leq M\|y_k - y_0\| + |\langle x_k, y_0 \rangle - \langle x_0, y_0 \rangle| \\ &< \epsilon, \end{aligned}$$

for all $k \geq K$, which completes the proof. □

10. Let X be a B^* space. Show that any closed convex set E in X is weakly closed. (E is called *weakly closed* if $x_0 \in E$ whenever $\{x_k\} \subset E$ and $x_k \rightarrow x_0$.)

Proof. Let $\{x_k\} \subset E$ and $x_k \rightarrow x_0$, then there exists convex combinations $y_k = \sum_{i=1}^k \lambda_i^k x_i$ of $\{x_k\}$ such that $y_k \rightarrow x_0$. Since $\{y_k\} \subset E$ and E is closed, we know $x_0 \in E$. □